

Weight Tying Causes Grokking in Conditional Flow Matching

Kosuke Ukita^{*1} Tsuyoshi Okita^{*2}

^{*}Department of Artificial Intelligence, Kyushu Institute of Technology, Iizuka, Fukuoka, Japan
¹ukita.kosuke299@mail.kyutech.jp, ²tsuyoshi@ai.kyutech.ac.jp

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Introduction: Grokking is delayed generalization: a network fits its training set and then, long after a plateau, suddenly begins to fit unseen data. It was reported for transformers solving algebraic tasks over symbol strings (Power et al. 2022), then for image classification (Liu et al. 2023), and recently for a diffusion model solving modular addition rendered as images (Kim et al. 2026). Factors reported to drive the delay include the amount of training data, optimization, the initialization scale, and adjustments to normalization and attention (Power et al. 2022; Liu et al. 2023; Yildirim 2026). All of these are reports in a classification setting, including token prediction over symbol strings. This work answers the question of whether a structural choice specific to conditional generative models acts on delayed generalization.

Approach: Each of the 23 symbols is drawn as one 32×32 grayscale image. The model receives two symbol images a, b as conditioning and generates the image of their sum $c = a + b \pmod{23}$ from pure noise in a single step; the generated image is decoded to the nearest of the 23 prototypes and scored (Fig. 1). Of the 276 unordered pairs, only 248 are used for training and the remaining 28 are held out; generalization is measured on these unseen pairs alone. The same rule is also implemented as $16 \times 16 \times 8$ -frame synthetic video.

Our conditional flow matching takes the noisy target x_t together with the conditioning images a, b . We compare a Tied design, in which a single linear embedding encodes all three, against an Untied design that gives the conditioning inputs and the target separate embedding layers. We further set the target embedding to $z_t = \cos \theta \cdot W_c x_t + \sin \theta \cdot W_g x_t$, so that the path between Tied ($\cos \theta = 1$) and Untied ($\cos \theta = 0$) can be traversed continuously, and we evaluate that trajectory as well; because the mixture is orthogonal, the initialization variance and the parameter count are identical at every level. Nothing else changes.

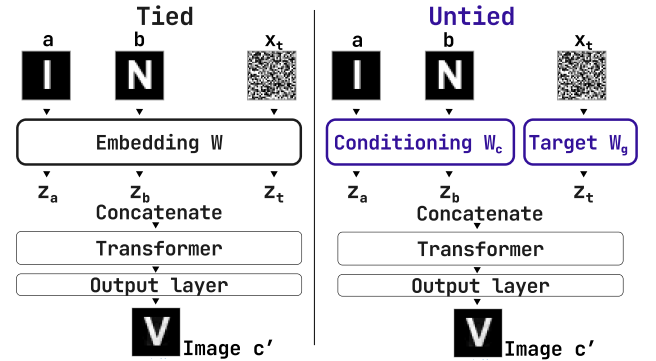
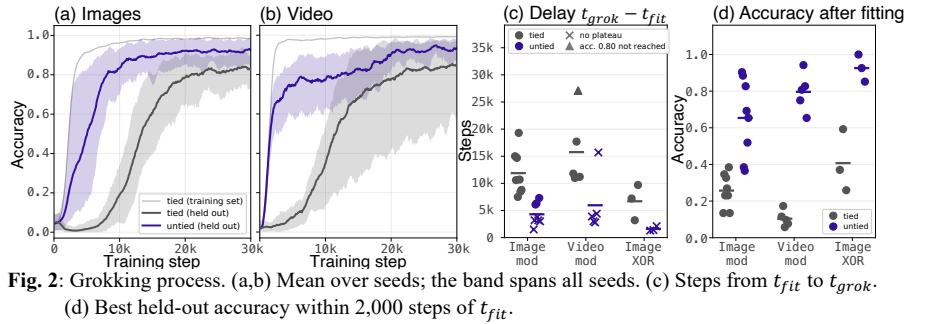


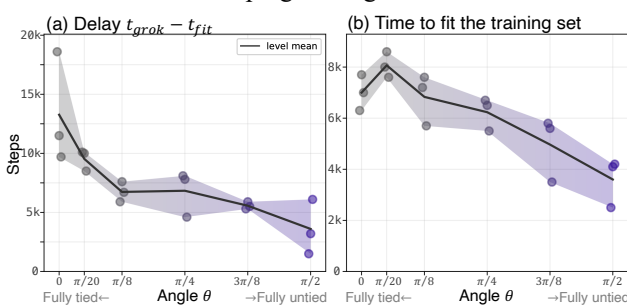
Fig. 1: Tied encodes the conditioning images a, b and the noisy target x_t with one embedding; Untied gives them separate embeddings.

Results and Discussion:

Tied meets the grokking rule in 15 of 16 runs, Untied in only 3 of 16. Immediately after the training set has been memorized, held-out accuracy is 0.10–0.41 for Tied against 0.65–0.93 for Untied, and Untied is the higher of the two in 96 of 98 pairwise comparisons



(Fig. 2(d)): Tied has a memorization phase to escape, whereas Untied generalizes as soon as it fits. The delay is likewise 2.6–4.2 times longer (Fig. 2(c)). The effect holds across two modalities and two groups, and Untied has more parameters (+17% on images, +25% on video), so neither the task nor capacity explains it. Fig. 3 treats tying as a quantity rather than a condition: sweeping the angle θ from 0 to $\pi/2$ shortens the delay monotonically from 13,267 to 3,600 steps while



the parameter count is held identical at every level, so the delay tracks the amount of tying itself. Grokking here is therefore not a capability but a failure mode produced by one structural choice: in Tied a single encoder must read both the clean conditioning images and the noisy target, and the model appears to fall back on memorization until a representation serving both roles emerges. Untying this layer improves both speed and accuracy, and we propose encoder tying as one cause of the grokking delay in conditional generative models.